

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459339.6982 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1284 +/- 0.0081
Transit depth (Rp/Rs)^2	1.65 +/- 0.21 [%]
Orbital Inclination (inc)	88.04 +/- 0.76
Airmass coefficient 1 (a1)	0.97 +/- 0.009
Airmass coefficient 2 (a2)	0.028 +/- 0.021
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	#1 - [298, 450]
Optimal Aperture	4.22
Optimal Annulus	4.39
Transit Duration (day)	0.0943 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1284 $\pm$ 0.0081 vs 0.1063 (deviation 2.73 σ)
Duration fit vs published	0.0943 d vs 0.0867 d (deviation 1.82 σ)
Mid-time O-C	1.8 min
Depth SNR	15.9
Residual scatter	0.93 %

Light curve

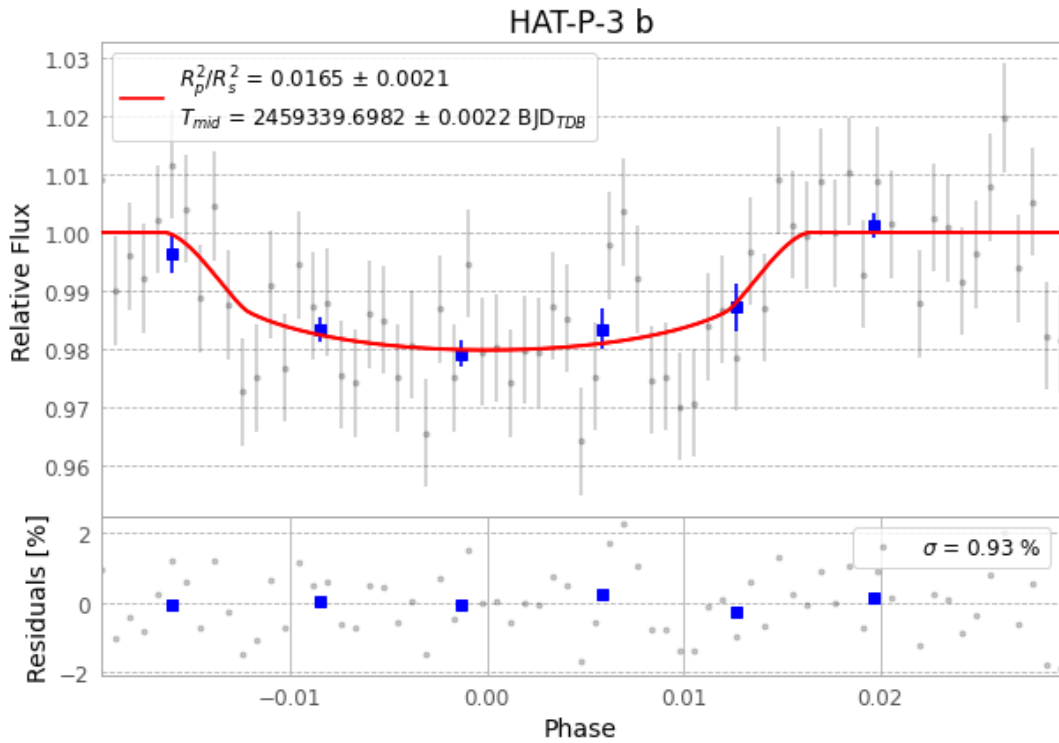


Figure 1 — FinalLightCurve_HAT-P-3 b_2021-05-04.png (SHA-256 e9a94057fb6dea46...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6ce4f997725eb143...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), HATP-3210505031812.FITS → HATP-3210505064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210505.log (d5ef0d35ddbe...), FinalLightCurve_HAT-P-3 b_2021-05-04.png (e9a94057fb6d...), FinalLightCurve_HAT-P-3 b_2021-05-04.csv (6a06b58464c5...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:01Z.